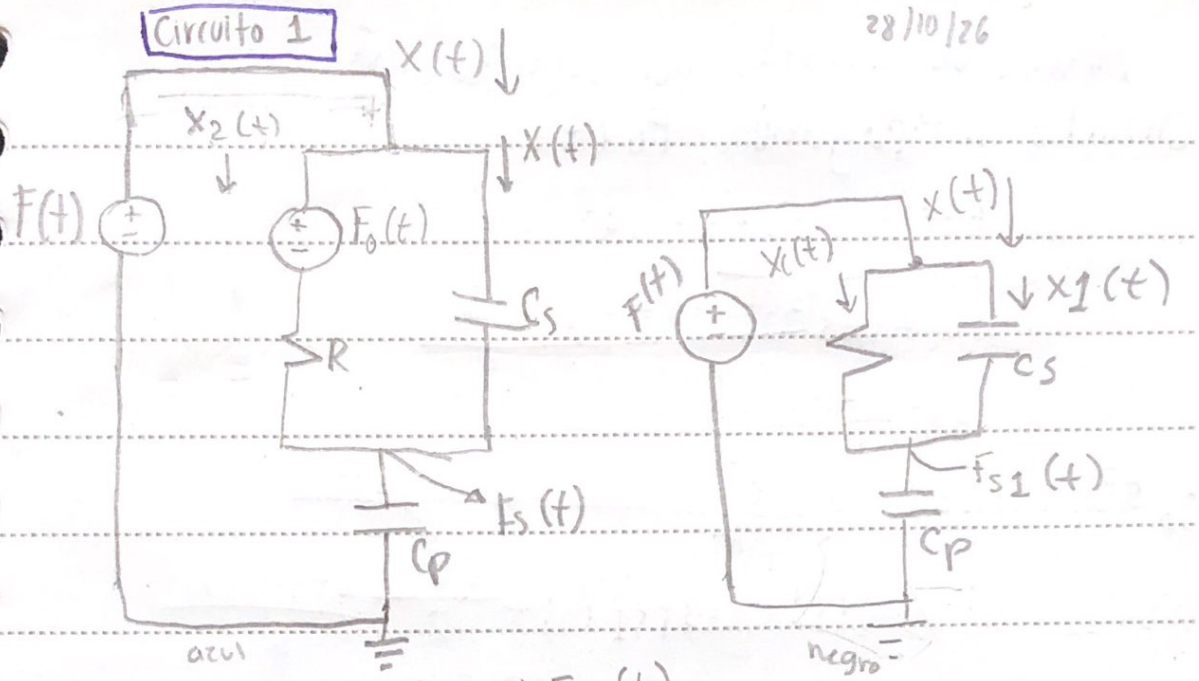
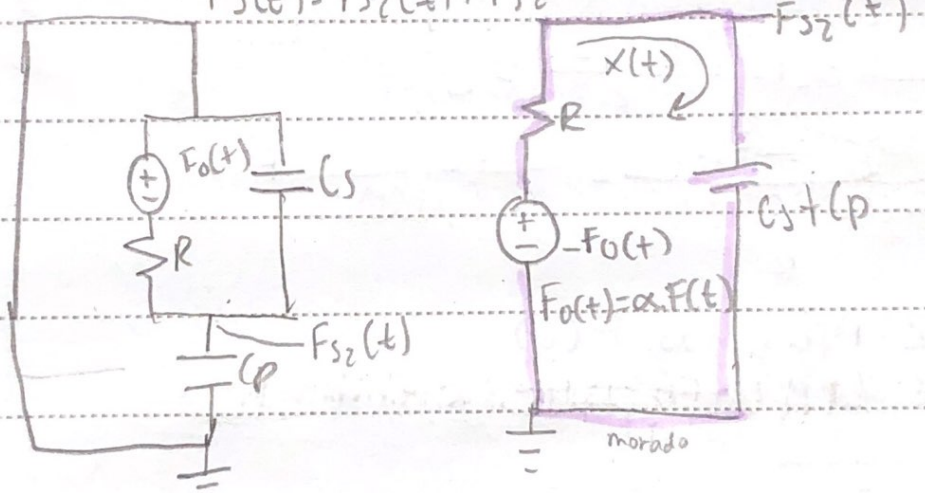
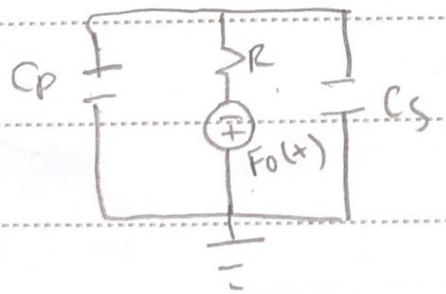
28/10/26

Circuito 1



$F_s(t) = F_{s2}(t) + F_{s2}(t)$





Analis del circuito por superposición

Circuito 1: Apagando $F_0(t)$

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R} F(s)}{\frac{R(C_p + C_s)s + 1}{R}} = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

Circuito 2: Apagando $F(t)$:

$$-aF(t) = Rx(t) + \frac{1}{C_p + C_s} \int x(t)\,dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s} \int x(t)\,dt$$

$$-aF(s) = RX(s) + \frac{X(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{X(s)}{(C_p + C_s)s} \rightarrow \text{salida}$$

$$F(s) = -\frac{X(s)}{a} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}$$

$$\frac{F_{s2}(s)}{F(s)} = \frac{\frac{X(s)}{(C_p + C_s)s}}{-\frac{X(s)}{a} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}} = -\frac{a}{R(C_p + C_s)s + 1}$$

$$F_{s2}(s) = -\frac{aF(s)}{R(C_p + C_s)s + 1}$$

Función de transferencia

$$F(s) = F_{s1}(s)\,F_{s2}(s) = \frac{C_s R_s + 1}{R(C_p + C_s)s + 1} F(s) - \frac{a F(s)}{R(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R_s + 1 - a}{R(C_p + C_s)s + 1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R_s + 1 - a}{R(C_p + C_s)s + 1}$$